

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

prathamesh Mandhane (16/09/26)



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Session	2026-27 (Odd)	Course Name	Lab: Neural Networks and Deep Learning
Semester	7 th	Course Code	23ADS1704
Roll No	A-52 (23070739)	Name of Student	prathamesh Mandhane

Practical Number	6
Course Outcome	CO1: Understand the fundamental concepts of deep learning and neural network architectures CO2: Apply different techniques to develop and optimize deep learning models. CO3: Analyze the performance of deep learning models using appropriate evaluation techniques. CO4: Design and implement deep learning solutions for real-world problems using modern tools and frameworks.
Aim	Implement an LSTM-based Autoencoder using Google Colab
Problem Definition	Implement an LSTM Autoencoder for IoT machine temperature data . The model should learn the normal pattern of temperature sequences and reconstruct them. A high reconstruction error indicates that the sequence is different from the learned normal pattern and can therefore be considered an anomaly
Theory (100 words)	An LSTM Autoencoder is a neural network that learns to encode a sequence into a compact representation and then decode it to reconstruct the original sequence. LSTM is suitable for time-series data because it can learn dependencies between observations over time. The encoder compresses the input sequence into a fixed-size representation, while the decoder reconstructs the original sequence. The model is trained using Mean Squared Error (MSE) between the input and reconstructed output. After training, reconstruction error is calculated for each sequence. If the error is greater than a selected threshold, the sequence is classified as an anomaly. This technique is useful for IoT monitoring, machine fault detection, and sensor anomaly detection. .



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Procedure and Execution (100 Words)	Algorithm: 1. Import NumPy, Pandas, Matplotlib, Scikit-learn and TensorFlow/Keras libraries. 2. Generate synthetic IoT temperature data. 3. Add abnormal temperature values to create anomalies. 4. Store the data in a DataFrame. 5. Normalize the temperature values using MinMaxScaler. 6. Divide the time-series data into sequences of 30 time steps. 7. Create the LSTM Autoencoder. 8. Use an LSTM layer as the encoder. 9. Use RepeatVector to repeat the encoded representation. 10. Use another LSTM layer as the decoder. 11. Reconstruct the input using TimeDistributed(Dense()). 12. Compile the model using Adam optimizer and MSE loss. 13. Train the model using the input sequences as both input and target. 14. Calculate reconstruction error for every sequence. 15. Calculate an anomaly threshold using mean + 3 × standard deviation. 16. Classify sequences with error greater than the threshold as anomalies. 17. Visualize reconstruction error and the anomaly threshold.
	<pre>import numpy as np import pandas as pd import matplotlib.pyplot as plt from sklearn.preprocessing import MinMaxScaler from tensorflow.keras.models import Model from tensorflow.keras.layers import Input, LSTM, RepeatVector from tensorflow.keras.layers import TimeDistributed, Dense</pre>



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Step 2: Generate IoT Temperature Data

```
np.random.seed(42)
```

```
# Generate normal sensor data
```

```
time = np.arange(1000)
```

```
temperature = (
```

```
    50
```

```
    + 2 * np.sin(time / 20)
```

```
    + np.random.normal(0, 0.5, 1000)
```

```
)
```

```
# Add abnormal/anomalous values
```

```
temperature[700:710] += 10
```

```
temperature[850:860] -= 8
```

```
# Create DataFrame
```

```
data = pd.DataFrame({
```

```
    'Temperature': temperature
```

```
})
```

```
print(data.head())
```

Output

```
Temperature
```

```
0    50.248357
```

```
1    50.030826
```

```
2    50.523511
```

```
3    51.060391
```

```
4    50.280262
```

The first five temperature values are approximately **50°C**, representing the normal sensor readings.

Step 3: Visualize Temperature Data

```
plt.figure(figsize=(12, 4))
```

```
plt.plot(data['Temperature'])
```



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```
plt.title("IoT Machine Temperature Data")
plt.xlabel("Time")
plt.ylabel("Temperature")
```

```
plt.show()
```

Output

A time-series graph is displayed. The abnormal temperature regions can be observed as sudden increases and decreases around the artificially introduced anomaly regions.

Step 4: Normalize the Data

```
scaler = MinMaxScaler()
```

```
scaled_data = scaler.fit_transform(
    data[['Temperature']]
)
```

```
print(scaled_data[:5])
```

Output

```
[[0.53071978]
 [0.51977902]
 [0.54455870]
 [0.57156118]
 [0.53232444]]
```

Normalization converts the temperature values into a range suitable for neural-network training.

Step 5: Create Time-Series Sequences

```
sequence_length = 30
```

```
X = []
```

```
for i in range(len(scaled_data) - sequence_length):
    X.append(
```



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```
scaled_data[i:i + sequence_length]
```

```
)
```

```
X = np.array(X)
```

```
print("Shape of input sequences:", X.shape)
```

Output

Shape of input sequences: (970, 30, 1)

Explanation

The shape is:

970 → Number of sequences

30 → Time steps in each sequence

1 → Number of features

Therefore, each input to the LSTM contains **30 consecutive temperature readings**.

Step 6: Build the LSTM Autoencoder

```
timesteps = X.shape[1]
```

```
features = X.shape[2]
```

```
# Input layer
```

```
inputs = Input(  
    shape=(timesteps, features)  
)
```

```
# Encoder
```

```
encoded = LSTM(  
    64,  
    activation='relu'  
) (inputs)
```

```
# Repeat encoded representation
```

```
decoded = RepeatVector(  
    timesteps  
) (encoded)
```



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	<pre># Decoder decoded = LSTM(64, activation='relu', return_sequences=True)(decoded) # Output layer outputs = TimeDistributed(Dense(features))(decoded) # Create model autoencoder = Model(inputs, outputs) autoencoder.summary()</pre> Output: Reconstruction Loss (MSE): 0.0034712827764451504 Anomaly Threshold: 0.03038638059457798 Total Anomalies Detected: 39 These values are from the provided Colab execution; exact values can vary slightly between runs.
Output Analysis	• The LSTM-based Autoencoder was successfully implemented in Google Colab. It learned the normal temperature patterns and detected abnormal sequences using reconstruction error. The obtained reconstruction MSE was approximately 0.00347, with an anomaly threshold of approximately 0.03039, resulting in 39 detected anomalous sequences in the provided run.



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Link of student Github profile where lab assignment has been uploaded	https://github.com/Prath318/NNDL-Practical
Conclusion	The LSTM Autoencoder successfully learned the temporal patterns of IoT temperature data and reconstructed normal sequences with low error. Reconstruction error was used as the evaluation measure instead of classification accuracy because the Autoencoder performs unsupervised anomaly detection . Sequences having reconstruction error greater than the calculated threshold were identified as anomalies. Thus, LSTM Autoencoders can be effectively used for time-series anomaly detection in applications such as IoT sensors, machine monitoring, and predictive maintenance.
Date	16/09/26